Appendix A: Detailed Proofs

This appendix provides detailed proofs of the theoretical properties discussed in Section 4.2.5, focusing on the bias and variance of softmax-based and Dirichlet-based policies for simplex-constrained action spaces.

A.1 Bias of Softmax-based Policy

Proposition 1 (Bias of Softmax-based Policy)**.** *A softmax-based policy that maps Gaussian logits to a probability simplex introduces estimation bias due to the translation-invariance property $\sigma(\mathbf{z} + c\mathbf{1}) = \sigma(\mathbf{z})$. The resulting policy density marginalizes over an unbounded latent shift c, which is ignored by standard estimators.*

Proof. Let $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \sigma^2 I)$ and define $a_i = \exp(z_i)/\sum_{j=1}^{k} \exp(z_j)$. For any $c \in \mathbb{R}$,

$$\sigma(\mathbf{z} + c\mathbf{1}) = \sigma(\mathbf{z}),$$

so the preimage of any $\mathbf{a}$ is an affine 1-D set $\{\mathbf{z} + c\mathbf{1} \mid c \in \mathbb{R}\}$. Consequently, the action density is

$$\pi(\mathbf{a} \mid \boldsymbol{\mu}) = \int_{-\infty}^{+\infty} p(\mathbf{z} + c\mathbf{1} \mid \boldsymbol{\mu}, \sigma^2 I) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{a}} \right) \right| dc,$$

which explicitly integrates over the latent shift c. The true gradient $\nabla_{\boldsymbol{\mu}} J$ should differentiate this marginalized density. However, the standard REINFORCE estimator samples a single $\mathbf{z}$ (without modeling c) and uses $\nabla_{\boldsymbol{\mu}} \log p(\mathbf{z} \mid \boldsymbol{\mu})$ as if the mapping were injective. Thus,

$$\mathbb{E}_{\mathbf{z}}[\nabla_{\boldsymbol{\mu}} \log \pi(\mathbf{a} \mid \mathbf{z})\, A(s, \mathbf{a})] \neq \nabla_{\boldsymbol{\mu}} J(\boldsymbol{\mu}),$$

establishing the bias. □

A.2 Variance Blow-up in Softmax-based Policy

Proposition 2 (Variance Blow-up in Softmax-based Policy)**.** *For the softmax-based policy, the Fisher information matrix scales inversely with the variance of the underlying Gaussian logits: $F(\boldsymbol{\mu}) \propto I/\sigma^2$. As $\sigma \to 0$, the variance of the score-function estimator diverges, destabilizing learning.*

Proof. The log-density of $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \sigma^2 I)$ is

$$\log p(\mathbf{z} \mid \boldsymbol{\mu}) = -\frac{1}{2\sigma^2} \|\mathbf{z} - \boldsymbol{\mu}\|^2 + \text{const},$$

thus

$$\nabla_{\boldsymbol{\mu}} \log p(\mathbf{z} \mid \boldsymbol{\mu}) = \frac{\mathbf{z} - \boldsymbol{\mu}}{\sigma^2}.$$

The Fisher information w.r.t. $\boldsymbol{\mu}$ is

$$F(\boldsymbol{\mu}) = \mathbb{E}_{\mathbf{z}}\big[(\mathbf{z} - \boldsymbol{\mu})(\mathbf{z} - \boldsymbol{\mu})^\top\big]/\sigma^4 = \sigma^2 I/\sigma^4 = I/\sigma^2.$$

Therefore, as $\sigma \to 0$, $F(\boldsymbol{\mu}) \to \infty$. Without compensating for the step size (e.g., by natural gradient), the variance of the estimator, and thus parameter updates, can explode. □

A.3 Unbiasedness of Dirichlet-based Policy

Proposition 3 (Unbiasedness of Dirichlet-based Policy)**.** *A Dirichlet-based policy samples $A \sim \mathrm{Dir}(\boldsymbol{\alpha})$, which inherently satisfies the simplex constraint. The policy gradient estimator remains unbiased.*

Proof. The Dirichlet density is

$$p(\mathbf{a} \mid \boldsymbol{\alpha}) = \frac{\Gamma(\alpha_0)}{\prod_{j=1}^{k} \Gamma(\alpha_j)} \prod_{j=1}^{k} a_j^{\alpha_j - 1}, \quad \alpha_0 = \sum_{j=1}^{k} \alpha_j.$$

No additional latent shift or non-identifiable transformation is introduced. Therefore,

$$\nabla_{\boldsymbol{\alpha}} J(\boldsymbol{\alpha}) = \mathbb{E}_{\mathbf{a} \sim \mathrm{Dir}(\boldsymbol{\alpha})} \big[\nabla_{\boldsymbol{\alpha}} \log p(\mathbf{a} \mid \boldsymbol{\alpha}) A(s, \mathbf{a}) \big],$$

showing the REINFORCE estimator is unbiased. $\square$

A.4 Variance Control in Dirichlet-based Policy

Proposition 4 (Variance Control in Dirichlet-based Policy)**.** *For $\mathbf{a} \sim \mathrm{Dir}(\boldsymbol{\alpha})$,*

$$\mathrm{Var}[a_j] = \frac{\alpha_j(\alpha_0 - \alpha_j)}{\alpha_0^2(\alpha_0 + 1)}, \quad \alpha_0 = \sum_j \alpha_j.$$

As $\alpha_0 \to \infty$, the variance smoothly approaches 0, and the Fisher information remains bounded.

Proof. The variance formula is standard for the Dirichlet distribution. Let α_j/α_0 stay finite when $\alpha_0 \to \infty$; then

$$\mathrm{Var}[a_j] \sim \frac{1}{\alpha_0} \to 0.$$

Moreover, the second derivative of $\log p(\mathbf{a} \mid \boldsymbol{\alpha})$ involves the trigamma function $\psi'(\cdot)$, which vanishes as its argument grows, keeping the Fisher information bounded. Hence gradient variance remains controlled near determinism. $\square$

A.5 Remark on Boundary Expressiveness

Remark. Since $p(\mathbf{a} \mid \boldsymbol{\alpha}) \propto \prod_j a_j^{\alpha_j - 1}$, letting $\alpha_j \to 0^+$ increases the density near $a_j = 0$, enabling near-sparse solutions. By contrast, softmax outputs $a_j > 0$ for all j, and thus cannot realize exact zeros. This difference explains the superior expressiveness of Dirichlet policies for sparse routing.